Beyond Detection: Multimodal Mitigation of Hateful Memes

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Abstract

Hateful memes harm individuals and groups through a multimodal combination of image and text that bypasses single-modality moderation. We address the task of *mitigating* hateful memes while preserving their non-hateful content. Our two-stage pipeline uses a vision-language model to detect hate and localize its source, then a Stable Diffusion model guided by a prompt designed to modify as little of the image as possible while neutralising the hateful elements.

Keywords: Hate mitigation; Open-weights VLMs; Diffusion models; Finetuning

1. Introduction

Memes are one of the most widely used forms of online communication. While most are harmless, some convey hate against individuals or groups based on protected characteristics such as ethnicity, religion, or gender [1]. What makes hateful memes particularly challenging is their *multimodal* nature: the hateful meaning often emerges from the combination of image and text, even when each modality is benign in isolation.

Most prior work focuses on **detecting** hateful memes; far less attention has been paid to the task of **mitigating** them while preserving meaning. This work addresses the question: *how can we mitigate a hateful meme so that the hateful content is removed while its original meaning, humour and structure are preserved?* We propose a pipeline based on open-weights VLMs and diffusion model, under the hypothesis that recent advances in open-source models make them capable of meeting this challenge.

2. Related Work

Dataset. The Hateful Memes Challenge [1] defines hateful memes as multimodal attacks on protected characteristics and provides the standard benchmark dataset.

Hateful meme mitigation. Two recent systems serve as our main sources of inspiration. **UnHateMeme** [2] uses a

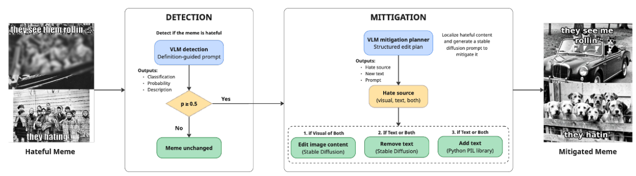

Figure 1. Suggested method: Detection and Mitigation that take as input a meme and output a mitigated version.

definition-guided VLM (GPT-4o) to detect hate, localize its source and replace the offending modality by rewriting the entire text with VLM or replacing the full image via top-k retrieval from an image corpus. **MemeDetoxNet** [3] localizes toxic visual regions with CLIP, applies global blurring and rewrites hateful captions with a LLM. It also introduces the *Toxicity Reduction vs. Context Preservation* trade-off.

Limitations of prior work. Both baselines attempt to preserve semantics but fall short. **MemeDetoxNet** applies a Gaussian blur to harmful regions, which obscures content without meaningfully altering it. **UnHateMeme** replaces the image via top-k similarity matching, substituting the original with a retrieved image rather than editing it. More broadly, no dedicated dataset exists for the mitigation task, making quantitative evaluation of semantic preservation an open challenge. Our approach uses prompt-guided image editing designed to minimise visual changes while neutralising hateful content.

3. Method

The pipeline consists of two stages, as illustrated in Fig 1.

Detection. Based on a hate definition, a VLM classifies each meme as hateful or not, returning a binary label, a hate probability and a natural-language description of the harmful content. Memes below a probability threshold ($p = 0.5$) are passed through unchanged.

Mitigation. For flagged memes, a VLM localizes the hate source into one of four categories (visual, textual, combined, or intersectional) and produces a structured plan: an image-editing prompt designed to minimise visual changes while neutralising the harmful visual content, alongside a safe replacement for the hateful text. A diffusion model then edits the image from this prompt, and the original text overlay is

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erased and redrawn with the replacement text.

Once the pipeline is deployed on the cluster, our comparison strategy keeps the diffusion model fixed and varies only the VLM component. The diffusion model offers limited independent leverage: since its input prompt is generated by the VLM, improving the VLM directly improves the diffusion output. Fine-tuning the diffusion model itself is also ruled out by the absence of paired training data, no dataset of (*hateful meme, mitigated meme*) pairs exists at sufficient scale. We evaluate three strategies of increasing complexity: the **baseline** prompts the VLM with only the hate speech definition and no examples; **prompt tuning** augments the prompt with category-labeled examples to steer the model's reasoning; **fine-tuning** trains the VLM directly on task-specific data.

3.1. Prompting Strategy Exploration

We explore three directions for improving detection through prompt design. First, **few-shot examples**: we provide the model with labeled meme examples before asking it to classify, using either synthetic examples designed to cover diverse hate types or real annotated examples from the dataset. Second, **affective conditioning**: we ask the model to first assess the tone of the meme (humour, sarcasm, offensiveness) before classifying it as hateful, either in a single prompt or as two sequential calls. Third, **thematic routing**: we first assign each meme to a category (historical, cultural, or identity-based) and then apply category-specific prompting, on the hypothesis that hate takes different forms depending on the theme.

3.2. Fine-tuning

Classification head. Our first fine-tuning attempt applied LoRA directly to a VLM using only the binary labels (hateful / not hateful) from the Facebook Hateful Memes dataset. This yielded catastrophic results: the model lost its ability to reason over meme content and collapsed to near-trivial predictions. Training on a binary signal alone provides no supervision on *why* a meme is hateful, stripping the model of the multimodal reasoning it needs. This motivated a lighter-weight alternative: the VLM is kept fully frozen and a single linear classification layer is trained on top of the last-token hidden state from the language model trunk, preserving the pretrained representations while adapting only the decision boundary.

Synthetic data generation. To recover the benefits of LoRA fine-tuning without the supervision collapse, we generate richer training data from the Hateful Memes dataset using Claude Sonnet 4.6 as a teacher model. The ground-truth binary label is provided in the generation prompt, anchoring Claude's outputs to the dataset's human annotations. For each meme, Claude produces a structured annotation covering hate probability, a natural-language explanation, hate type and localisation category and a full mitigation plan. Eval-set memes are excluded to prevent leakage. This yields 2,220 samples (1,499 hateful / 721 non-hateful), formatted into a **detection dataset** of 2,220 samples and a **mitigation dataset** of 1,499 hateful-only samples. Annotation quality is validated by manual inspection of a random subset.

LoRA adapter training. LoRA adapters are applied to all linear layers of the language model trunk, leaving the vision tower frozen. Freezing the vision tower preserves the model's pretrained visual understanding, avoiding the same collapse observed with naive LoRA fine-tuning where the model lost its ability to reason over meme content. We train two separate adapters, one for detection and one for mitigation planning, each supervised on the output format it will produce at inference time.

4. Validation

Model selection. For the VLM, we select 3 open-weights models that support image inputs and can run on the cluster. *google/gemma-4-31B-it*[1] and *Qwen/Qwen3.6-27B*[2] are two large models of comparable scale from different providers, allowing us to assess whether performance differences are model-specific or provider-agnostic. *Qwen/Qwen2.5-VL-7B-Instruct*[3] is a smaller model included to study the effect of scale on detection and planning quality. For the diffusion step, we use *black-forest-labs/FLUX.2-klein-9B*[4], an open-weights model that fits within the cluster's GPU memory constraints and achieves state-of-the-art image generation quality among models of its size.

Metrics. For detection, we report AUROC, macro-F1 and recall on a balanced 490-meme sample (245 hateful / 245 non-hateful). For mitigation, we measure toxicity reduction via TR% (share of memes judged non-hateful after editing) and content preservation via BERTScore (text similarity), CLIPScore (image-text coherence) and SSIM (visual fidelity to the original).

Pipeline	AUROC	F1	Recall
`fewshot_synthetic`	**0.755**	**0.710**	0.710
`fewshot_real`	0.749	0.672	0.829
`category_fewshot`	0.721	0.662	**0.857**
`category_sentiment`	0.661	0.646	0.702
`sentiment_single`	0.648	0.624	0.637
`sentiment_chained`	0.634	0.612	0.653

Table 1. Prompting pipeline comparison on a balanced 490-meme evaluation set. Metrics reported for the hateful class.

[1] https://huggingface.co/google/gemma-4-31B-it

[2] https://huggingface.co/Qwen/Qwen3.6-27B

[3] https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct

[4] https://huggingface.co/black-forest-labs/FLUX.2-klein-9B

Prompting strategy comparison. Results are reported in Tab. 1. On Qwen3.6, few-shot approaches outperform affective conditioning across all metrics, suggesting that concrete labeled examples are more informative than affective labels for this task. `fewshot_synthetic` achieves the best AUROC (0.755) and Macro-F1 (0.710), while few-shot pipelines using real examples trade precision for higher recall. Affective conditioning consistently underperforms, with `sentiment_chained` being the weakest, likely due to error propagation across sequential calls. We select `category_fewshot` as the detection prompt for subsequent experiments: despite a slightly lower F1, its recall of 0.857 ensures that the mitigation stage operates on a more complete set of hateful memes.

Method	Model	AUROC	F1	Recall
Baseline	Qwen3.6	0.705 ± 0.004	0.651 ± 0.005	0.767 ± 0.004
	Qwen2.5	0.636 ± 0.003	0.570 ± 0.006	0.877 ± 0.004
	Gemma4	0.798 ± 0.005	**0.761 ± 0.004**	0.729 ± 0.007
F.S+Cat	Qwen3.6	0.734 ± 0.015	0.663 ± 0.009	0.856 ± 0.016
	Qwen2.5	0.731 ± 0.019	0.653 ± 0.011	0.739 ± 0.014
	Gemma4	0.746 ± 0.004	0.706 ± 0.003	0.735 ± 0.009
Cls. Head	Qwen3.6	**0.841 ± 0.008**	0.757± 0.007	0.759 ± 0.009
	Qwen2.5	0.741 ± 0.006	0.655 ± 0.005	0.661 ± 0.008
	Gemma4	0.827 ± 0.006	0.741 ± 0.005	0.741 ± 0.003
LoRA	Qwen3.6	0.799 ± 0.005	0.726 ± 0.009	0.727 ± 0.004
	Qwen2.5	0.714 ± 0.006	0.624 ± 0.004	0.874 ± 0.003
	Gemma4	0.816 ± 0.004	0.693 ± 0.005	**0.878 ± 0.008**

Table 2. Detection results across models and training regimes. Values are reported as mean ± standard error.

Detection results. Table 2 shows that Gemma4 is the strongest model overall, achieving the best AUROC and F1 in the baseline setting, but it benefits the least from additional training strategies. On smaller models, few-shot prompting and the classification head reliably improve over the zero-shot baseline, while results are more mixed on larger models. LoRA fine-tuning consistently boosts recall but does not always improve AUROC or F1, suggesting it shifts the decision threshold rather than improving the model's discriminative ability.

Method	Model	TR (%)	BERT	CLIP	SSIM
Baseline	Qwen3.6	43.40 ± 3.03	**0.948 ± 0.001**	0.300 ± 0.005	0.505 ± 0.002
	Qwen2.5	28.30 ± 2.85	0.87 ± 0.004	0.286 ± 0.003	**0.543 ± 0.002**
	Gemma4	**83.85 ± 1.97**	0.930 ± 0.001	0.294 ± 0.005	0.510 ± 0.003
LoRA	Qwen3.6	63.28 ± 1.81	0.920 ± 0.001	**0.304 ± 0.002**	0.300 ± 0.002
	Qwen2.5	22.82 ± 2.10	0.921 ± 0.002	0.302 ± 0.003	0.297 ± 0.008
	Gemma4	51.21 ± 1.03	0.930 ± 0.001	0.303 ± 0.002	0.303 ± 0.005

Table 3. Mitigation results.

Mitigation results. Table 3 reveals a clear trade-off between toxicity reduction and visual preservation. Gemma4 vastly outperforms others models on toxicity reduction (84% vs. 63% for the second best one), while all models achieve similar scores on BERTScore and CLIPScore. BERTScore remains consistently high across all settings, indicating that text content is well preserved. CLIPScore values are uniformly low (≈0.30), but this should not be read as poor image-text coherence: CLIPScore was trained on natural image-caption pairs where the text describes the image, whereas in a meme the text is ironic commentary rather than a description, which intrinsically lowers the score. LoRA approximately halves SSIM compared to the baseline, likely because fine-tuned models generate more aggressive diffusion prompts that alter larger portions of the image.

Limitations of mitigation evaluation. The automatic metrics used capture only surface-level proxies of mitigation quality. BERTScore and SSIM measure similarity to the original, but a good mitigation should differ from the original by design. CLIPScore measures image-text coherence but not whether the hateful content was actually removed. None of these metrics can assess whether the mitigated meme retains its humour, structure or communicative intent. More faithful evaluation would require human annotation or an LLM-as-judge approach scoring toxicity reduction and semantic preservation jointly.

5. Conclusion

We presented a two-stage pipeline for hateful meme mitigation using open-weights VLMs and a diffusion model. Our results show that capable open-source models can detect and plan mitigation effectively, with Gemma4 achieving strong toxicity reduction without any fine-tuning. Few-shot prompting and classification head adaptation reliably improve detection on smaller models, while LoRA fine-tuning shifts the decision threshold toward higher recall at the cost of discrimination quality.

The main open challenge is evaluation: current automatic metrics capture surface-level similarity but cannot assess whether a meme has been genuinely neutralised while preserving its meaning. Future work should prioritise richer evaluation via LLM-as-judge or human annotation, as well as fine-tuning the diffusion model on meme-specific paired data to improve visual coherence of the mitigated outputs.

References

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